TEXAS A&M UNIVERSITY

Prototype: Counterfactual Student Intervention Dashboard

CSCE 679 Data Visualization

Yiming Xiao, Yi-Hsin Chiang

SELECT STUDENT

Focus Students

Student 1102 Critical
Actual 55 | Predicted 55.4

Student 5508 Critical
Actual 56 | Predicted 56.7

Student 5908 Critical
Actual 57 | Predicted 57.3

Student 5720 Critical
Actual 57 | Predicted 57.7

Student 2881 Critical
Actual 57 | Predicted 58.0

COMPARE BASELINE VS SCENARIO

Predicted Score Distribution

55
Actual exam score

OBSERVED BAND **Critical**
PASSING THRESHOLD **60**

55.4
Baseline expected score

90% INTERVAL **55.2 to 55.7**
RISK BELOW 60 **100%**

59.5
Scenario expected score

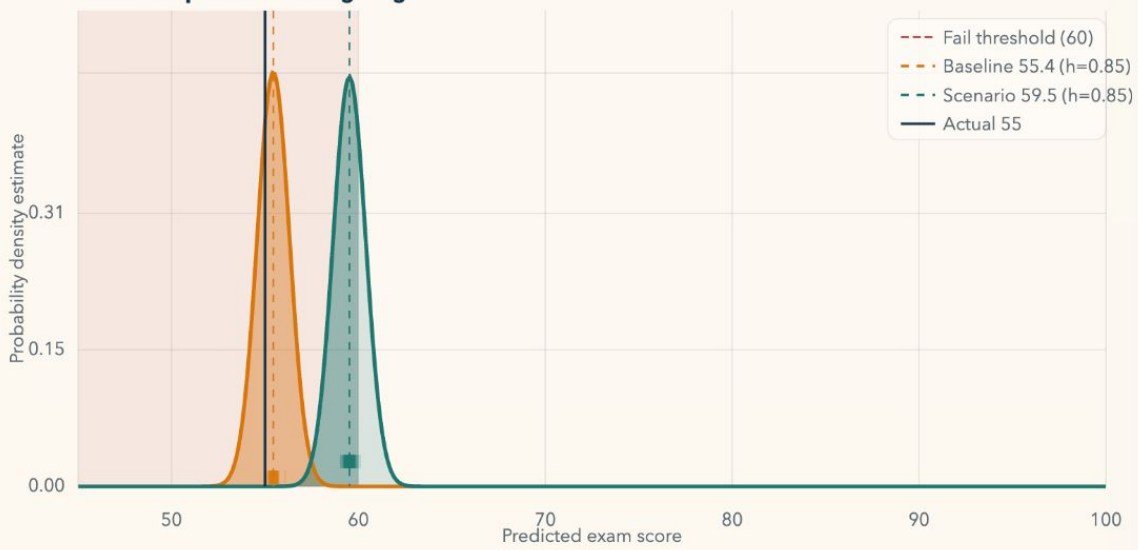
90% INTERVAL **59.2 to 59.8**
RISK BELOW 60 **71%**

+4.1
Lift from current intervention plan

FAIL-RISK CHANGE **-29 pts**
CHANGED LEVERS **3**

Applied recommendation: Tutoring sessions
Tutoring sessions moved from 6 /wk to 7 /wk. 3 levers now differ from baseline.

Red zone: predicted failing range below 60



DECISION SUPPORT

Best Next Levers

1. Access to resources +1.0
Next reasonable move: set **Access to resources** to **Medium**
Apply This Lever

2. Tutoring sessions +0.5
Next reasonable move: set **Tutoring sessions** to **8**
Apply This Lever

3. Hours studied +0.3
Next reasonable move: set **Hours studied** to **4**
Apply This Lever

4. Attendance +0.2
Next reasonable move: set

INTERVENTION CONTROLS

Editable Variables

Attendance recovery | Tutoring boost | Study plan | Reset to baseline

Attendance — Baseline
Directly actionable — **62%**
Baseline: 62%

Hours studied — Baseline
Directly actionable — **3 hrs**
Baseline: 3 hrs

Tutoring sessions — Edited
Directly actionable — **7 /wk**
Baseline: 1 /wk

Sleep hours — Baseline
Counseling or resource lever — **6 hrs**
Baseline: 6 hrs

Physical activity — Baseline
Counseling or resource lever — **3 days**
Baseline: 3 days

Motivation level — Edited
Counseling or resource lever — **High**
High
Baseline: Medium

Access to resources — Baseline
Counseling or resource lever — **Low**
Low
Baseline: Low

Extracurriculars — Edited
Counseling or resource lever — **Yes**
Yes
Baseline: No

Internet access — Baseline
Counseling or resource lever — **Yes**
Yes
Baseline: Yes

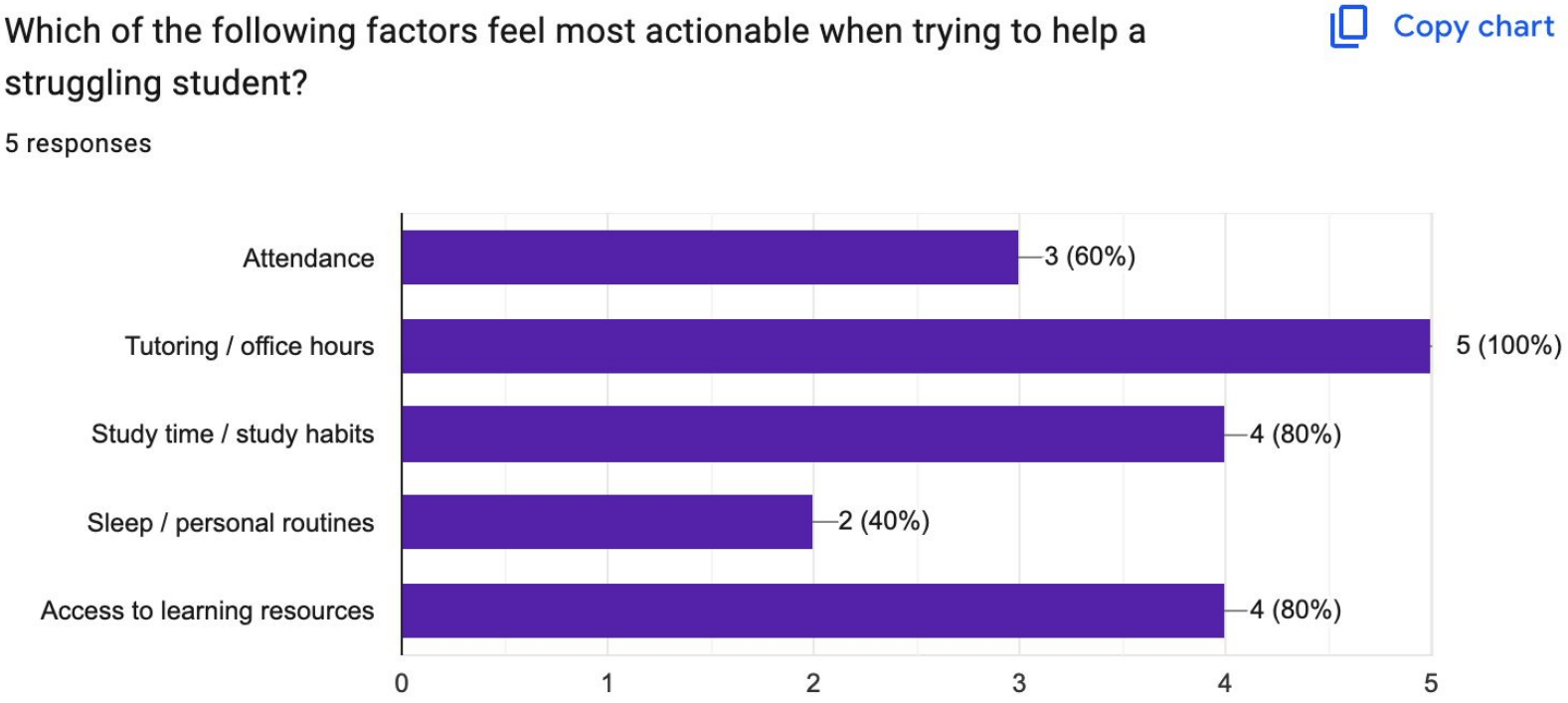
STUDENT CONTEXT

Immutable And Structural Factors

Previous score	67
Family income	Low
Teacher quality	Medium
Peer influence	Negative
Parental education	High School
Distance from home	Near
Gender	Male
School type	Public
Learning disabilities	No
Parental involvement	Medium

Interview (Questionnaire)

- 5 participants, including 2 professors and 3 TAs.



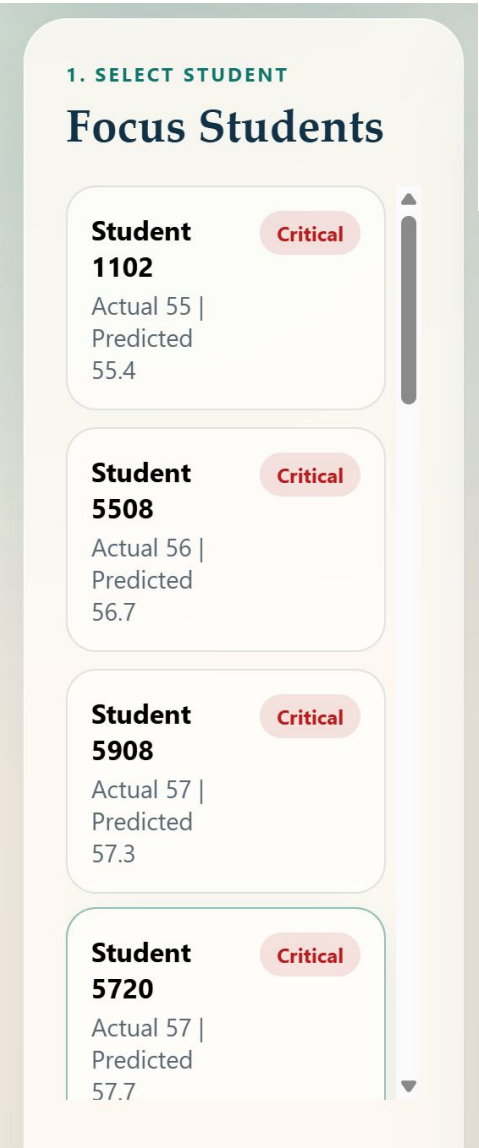
Design vs. Prototype Features

- D1: Quickly identify at-risk students
- D2: Compare baseline vs. intervention
- D3: Show interpretable guidance
- D4: Provide uncertainty and context

- F1: Focus Students panel
- F2: Predicted Score Distribution
- F3: Best Next Levers
- F4: Editable Intervention Controls
- F5: Context and Correlation Views

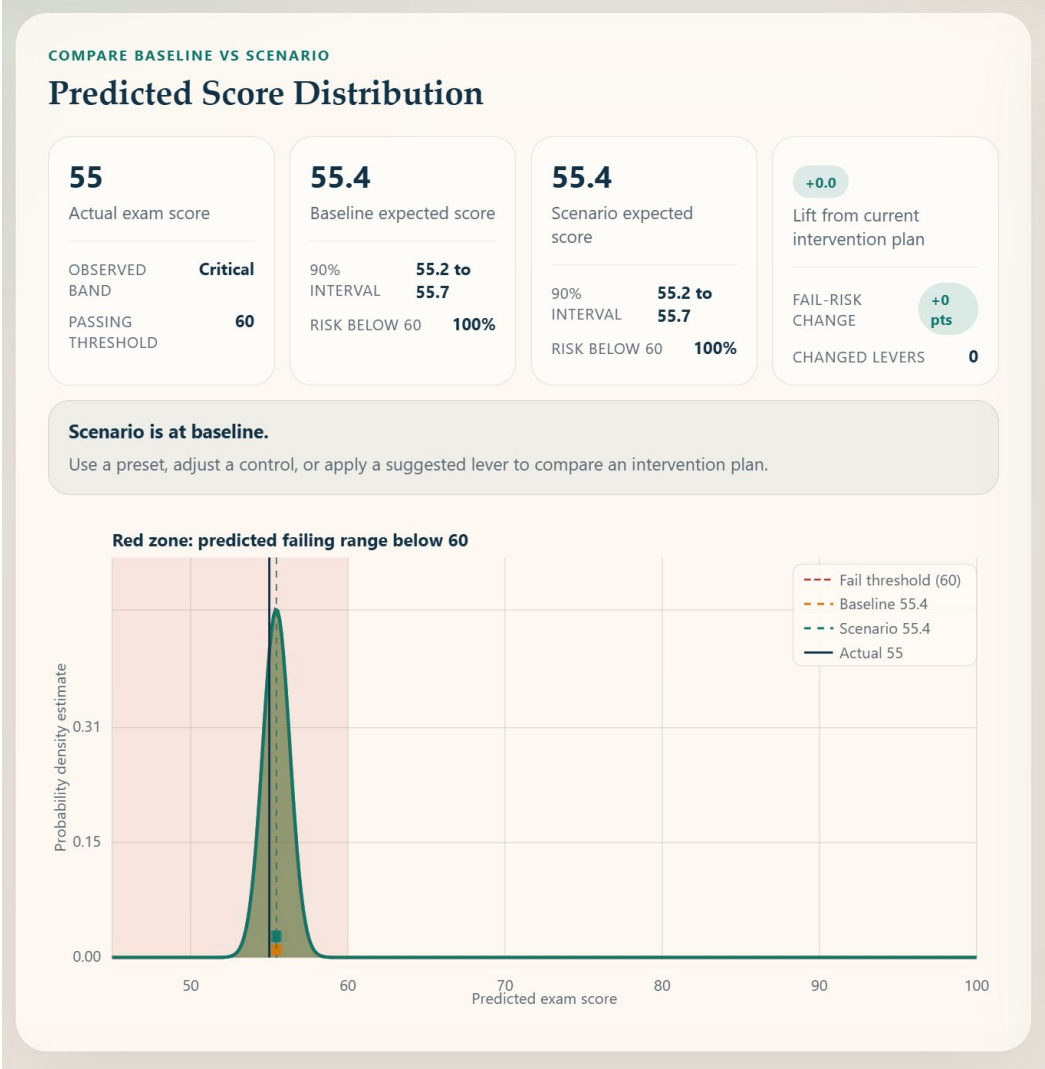
Overview

1. Select target student with critical score



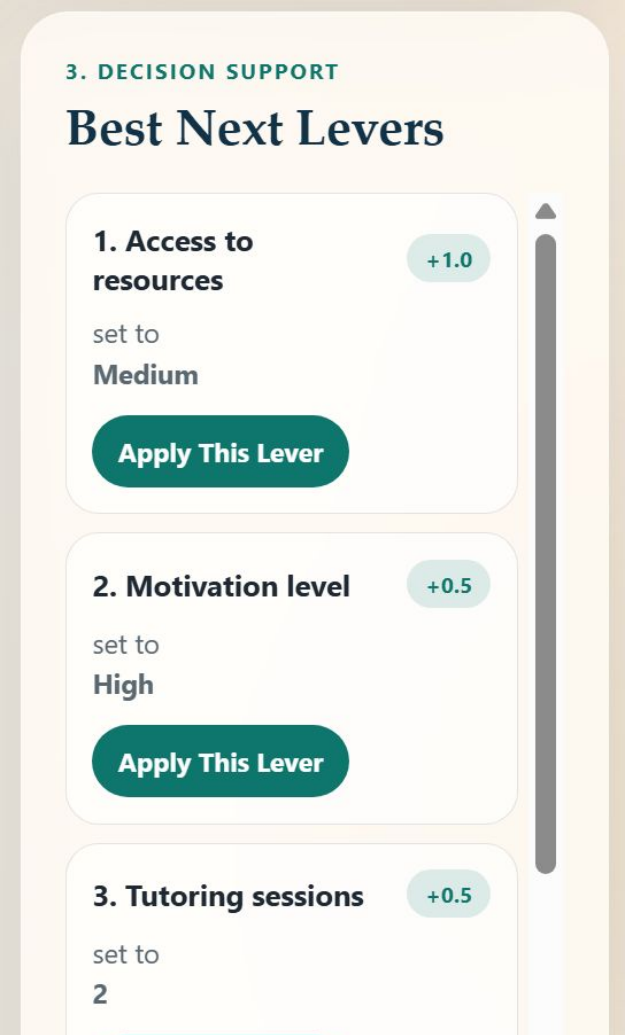
Overview

1. Select target student with critical score
2. Compare the actual score with model prediction



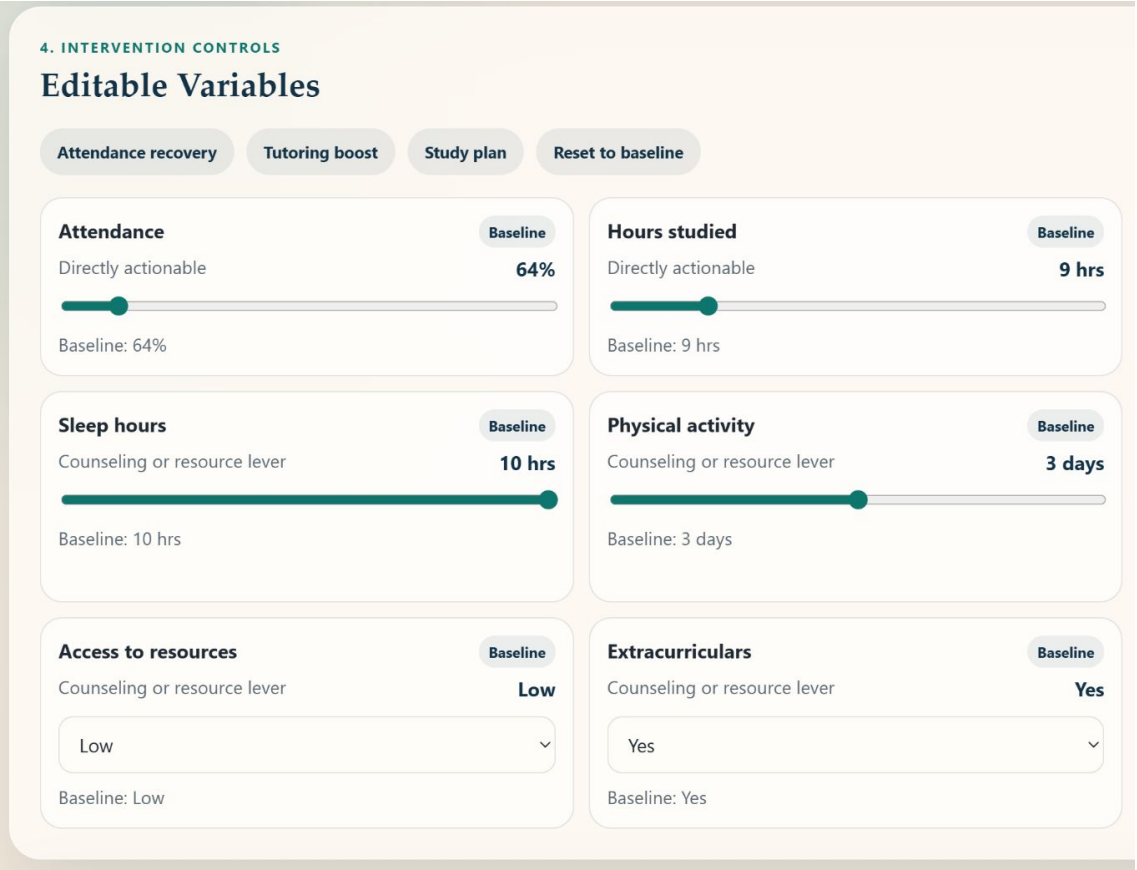
Overview

1. Select target student with critical score
2. Compare the actual score with model prediction
3. Apply suggested improvements



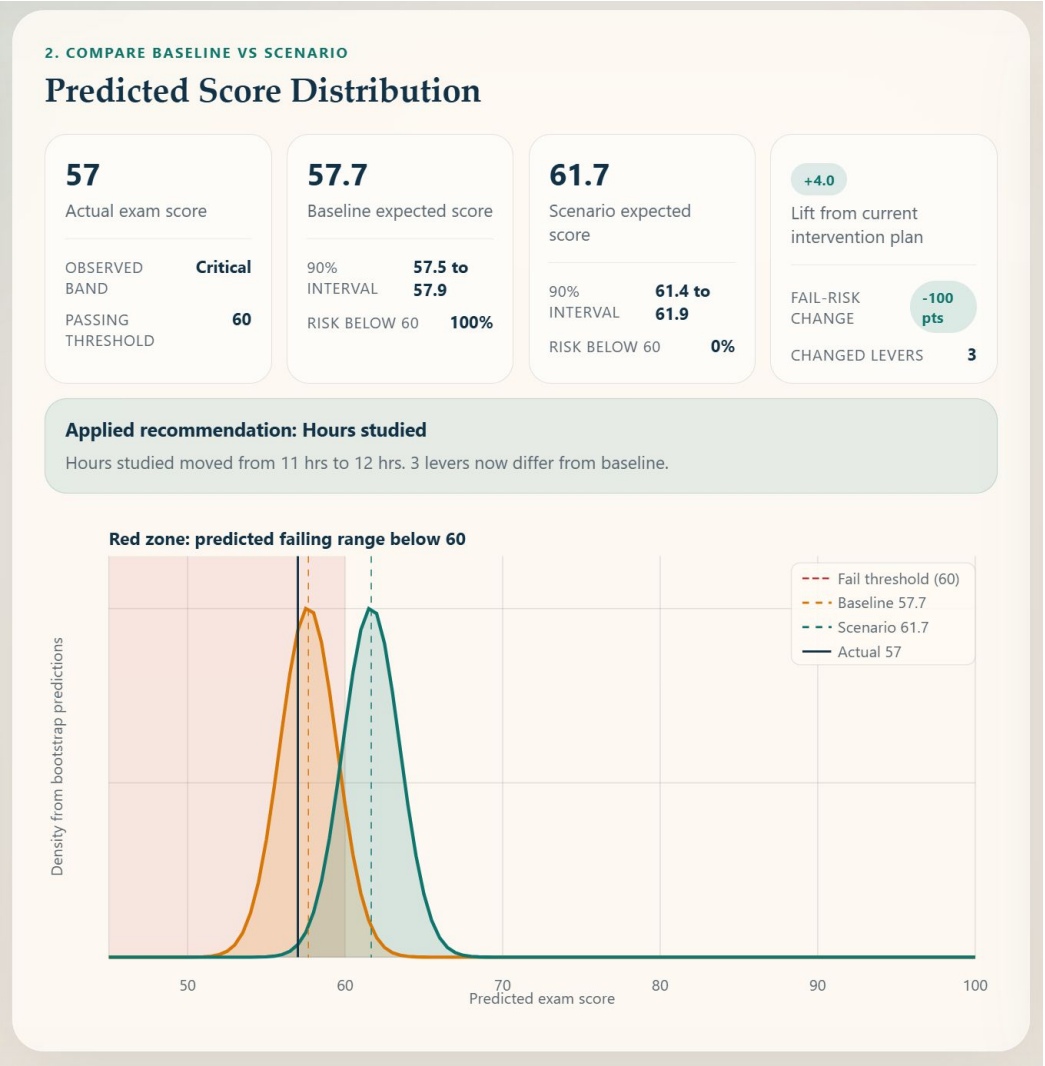
Overview

1. Select target student with critical score
2. Compare the actual score with model prediction
3. Apply suggested improvements
4. Manually tune the values for specified changes



Overview

1. Select target student with critical score
2. Compare the actual score with model prediction
3. Apply suggested improvements
4. Manually tune the values for specified changes
5. Observe the predicted score to improve under the adjustments



Overview

1. Select target student with critical score
2. Compare the actual score with model prediction
3. Apply suggested improvements
4. Manually tune the values for specified changes
5. Observe the predicted score to improve under the adjustments
6. Further confirm the immutable factors and the variables correlations with exam score

Demo

DATA VISUALIZATION PROTOTYPE

Counterfactual Student Intervention Studio

A prototype checkpoint for the advising dashboard proposed in the report and slides: choose an at-risk student, simulate an intervention, and see the predicted score distribution shift in real time.

6607
student records in the CSV

0.727
R-squared for the interpretable linear model

220
bootstrap models powering the uncertainty curves

1. SELECT STUDENT

Focus Students

Student 1102 Critical
Actual 55 | Predicted 55.4

Student Critical

2. COMPARE BASELINE VS SCENARIO

Predicted Score Distribution

57
Actual exam score

OBSERVED BAND **Critical**
PASSING THRESHOLD **60**

57.7
Baseline expected score

90% INTERVAL **57.5 to 57.9**
RISK BELOW **100%**

57.7
Scenario expected score

90% INTERVAL **57.5 to 57.9**
RISK BELOW **100%**

+0.0
Lift from current intervention plan

FAIL-RISK CHANGE +0 pts
CHANGED LEVERS **0**

3. DECISION SUPPORT

Best Next Levers

1. Access to resources +1.0
Next reasonable move: set **Access to resources** to **Medium**.